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# Diagnosing Search Ad Budget Waste: Segment Skew and Format–Device Mismatch in a Bicycle Retail Campaign

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KOI Research Agent  
ResearchOS

## Abstract

We analyze a May 2024 Yandex Direct search campaign for a regional bicycle retailer (Centra Market, Irkutsk) comprising 12,816 impressions, 1,391 clicks, and 50,005 RUB spend. Using a hypothesis-driven audit of campaign exports, we identify two structural drivers of inefficiency: (i) budget concentration in low-click-through-rate (CTR) ad groups, and (ii) a systematic format–device mismatch in which graphical ads on mobile achieve 3.32% CTR versus 10.65% for text ads on mobile—a 7.3 percentage-point gap that persists in all seven ad groups with paired formats. Counterfactual simulations on historical data show that reallocating 3,000 RUB from weak to strong groups yields only a modest +1.9% click gain, whereas disabling mobile graphical ads and reinvesting freed budget into mobile text ads recovers nearly all lost clicks while raising campaign CTR from 10.85% to 11.31%. We discuss implications for practitioner-facing campaign diagnostics and the limits of observational counterfactuals.

## 1 Introduction

Paid search advertising allocates limited budgets across thousands of keyword–audience–creative combinations. Practitioners routinely observe spend flowing into segments with poor engagement, yet isolating why budget is wasted—as opposed to merely where—requires structured hypothesis testing rather than dashboard inspection alone.

This paper presents a case study of search campaign efficiency for Centra Market, a bicycle retailer in Irkutsk, Russia. We work from a single-month export (`bicycle.csv`, May 2024) and pursue two complementary hypotheses:

1. Segment skew: A disproportionate share of budget flows to ad groups and query categories with below-median CTR, starving high-performing segments.
2. Format–device mismatch: Graphical creatives underperform on mobile relative to text creatives, wasting spend independent of keyword targeting.

For each hypothesis we specify measurable evidence criteria, run descriptive experiments on the export, and evaluate remediation policies via counterfactual budget simulations. Our contributions are methodological rather than algorithmic: we demonstrate how a small retail campaign can be diagnosed with transparent, reproducible analytics, and we report both positive and null-ish findings honestly—notably, group-level budget reallocation yields only modest gains, while format policy changes have a clearer efficiency signature.

## 2 Related Work

Search advertising optimization spans bid management [1], quality-score-aware budgeting [2], and creative A/B testing [3]. Device-specific performance heterogeneity is well documented in display and social advertising; mobile screen constraints and load latency plausibly amplify creative-format effects in search as well [4]. Our setting differs from large-scale auction simulations: we operate on a single advertiser’s row-level export with no access to auction logs, conversion labels, or live experimentation infrastructure. The analysis therefore resembles marketing mix post-mortems and observational policy evaluation more than reinforcement-learning bid optimizers [5].

## 3 Data and Experimental Protocol

### 3.1 Dataset

The campaign export contains 9,923 rows aggregated at the intersection of ad group, targeting category, ad format (text vs. graphical), and device type (mobile, desktop, tablet). Campaign-level totals for May 2024: 12,816 impressions, 1,391 clicks, 50,005 RUB spend, mean CTR  $\approx 10.85\%$ , mean cost-per-click (CPC) 35.9 RUB. There are 25 named ad groups and six targeting categories.

### 3.2 Metrics

We use standard search metrics:  $\text{CTR} = \text{clicks}/\text{impressions}$ ,  $\text{CPC} = \text{spend}/\text{clicks}$ , and clicks per 1,000 RUB spend ( $\text{cl}/1000\text{RUB}$ ) as a budget-efficiency index. Statistical tests are descriptive; we do not claim causal identification beyond counterfactual accounting on fixed historical prices.

### 3.3 Hypothesis Workflow

Each hypothesis decomposes into cause evidence (is the problem present?) and remediation (does a plausible fix help on paper?). Experiments are tracked on a kanban board; we report only completed cards with filled result tables.

## 4 Experiment 1: Segment-Level Budget Skew

### 4.1 Method

We group rows by ad group name and compute impressions, clicks, CTR, spend, CPC, and budget share. We flag “tail” segments with CTR below 8% and spend above 500 RUB. For remediation, we simulate transferring 1,000 RUB from each of three weak groups (Velosiped Katalog, Velosiped Stels, Velosiped Bolshoy) to three strong groups (Velosiped Detskiy, Velosiped Angarsk, Velosiped Podrostkovyy), holding total spend fixed and assuming marginal CPC in the recipient group.

### 4.2 Results

Table 1 contrasts exemplar high-spend groups. The top five groups by spend consume 60.3% of budget at an average CTR of 9.4%, below the campaign mean of 10.85%. Velosiped Katalog alone accounts for 9% of spend (4,566 RUB) at 8.39% CTR and 19.3  $\text{cl}/1000\text{RUB}$ , while Velosiped Detskiy spends 5.6% of budget (2,780 RUB) at 22.13% CTR and 29.9  $\text{cl}/1000\text{RUB}$ .

The 3,000 RUB reallocation simulation increases total clicks from 1,391 to 1,418 (+26.7 clicks, +1.9%). Pairwise transfers contribute +10.6 clicks (Katalog→Detskiy), +10.6 (Stels→Angarsk), and +5.5 (Bolshoy→Podrostkovyy). Campaign-wide  $\text{cl}/1000\text{RUB}$  rises from 27.8 to 28.4.

Table 1: Selected ad groups illustrating budget–efficiency skew (May 2024).

| Ad group          | Spend (RUB) | CTR (%) | Clicks | cl/1000RUB |
|-------------------|-------------|---------|--------|------------|
| Velosiped Katalog | 4,566       | 8.39    | 87     | 19.3       |
| Velosiped Detskiy | 2,780       | 22.13   | 83     | 29.9       |
| Campaign average  | –           | 10.85   | 1,391  | 27.8       |

### 4.3 Interpretation

Segment skew is confirmed: budget concentrates in broad, lower-CTR groups while niche high-intent groups remain underfunded. However, remediation is moderate: even a targeted 3,000 RUB shift—6% of monthly spend—lifts clicks by less than 2%. A backlog experiment (not executed) estimates that halving spend on broad/competitive/alternative query categories ( $\sim 1,467$  RUB) could yield  $\sim 40$  incremental clicks if reinvested at 36.6 RUB CPC; this remains prospective.

## 5 Experiment 2: Format–Device Interaction

### 5.1 Method

We cross-tabulate format and device over the full export, then repeat within ad groups where both text and graphical mobile creatives received  $\geq 20$  impressions. Success criteria: mobile graphical CTR at least 3 percentage points below mobile text CTR globally, and text winning in  $\geq 5/7$  paired groups.

For remediation we (a) remove the mobile graphical segment (542 impressions, 18 clicks, 556 RUB) and recompute campaign totals, then (b) reallocate the freed 556 RUB into mobile text at observed CPC 37.50 RUB.

### 5.2 Results

Table 2 summarizes the four main format–device cells. Mobile graphical ads achieve 3.32% CTR versus 10.65% for mobile text ( $\Delta = 7.3$  pp). Desktop graphical ads perform at 12.85% CTR, comparable to the campaign average. Mobile text carries 9,229 impressions and 983 clicks (36,863 RUB).

Table 2: CTR by ad format and device type (May 2024).

| Format    | Device  | Impr. | Clicks | CTR (%) | CPC (RUB) |
|-----------|---------|-------|--------|---------|-----------|
| Text      | Mobile  | 9,229 | 983    | 10.65   | 37.50     |
| Graphical | Mobile  | 542   | 18     | 3.32    | 30.87     |
| Text      | Desktop | 14    | 4      | 28.57   | 34.50     |
| Graphical | Desktop | 2,826 | 363    | 12.85   | 32.59     |

In paired ad-group comparisons, text beats graphical on mobile in 7/7 groups (gaps +1.0 to +11.0 pp). Examples: Velosiped Kupit 9.31% vs. 2.70%; Velosiped Angarsk 14.35% vs. 3.33%.

Policy simulation results appear in Table 3. Removing mobile graphical ads raises CTR from 10.85% to 11.19% but costs 18 clicks. Reinvesting 556 RUB at mobile-text CPC yields +14.8 expected clicks, netting 1,388 vs. 1,391 baseline ( $-3$  clicks,  $-0.2\%$ ) while lifting CTR to 11.31%.

### 5.3 Interpretation

The format–device mismatch hypothesis is strongly supported. Weakness is localized to graphical $\times$ mobile, not graphical ads globally nor mobile traffic generally. Paired compar-

Table 3: Counterfactual format policy on historical May data.

| Scenario                    | Clicks | CTR (%) | Spend (RUB) |
|-----------------------------|--------|---------|-------------|
| Observed (May)              | 1,391  | 10.85   | 50,005      |
| Mobile graphical off        | 1,373  | 11.19   | 49,450      |
| Off + budget to mobile text | 1,388  | 11.31   | ~50,006     |

isons rule out the confound that graphical creatives were merely assigned to intrinsically weak keywords.

## 6 Discussion and Limitations

Comparing remedies. Segment reallocation and format policy operate on different levers. Our simulations suggest format policy delivers a clearer CTR efficiency gain per ruble of identified waste (556 RUB at 3.32% CTR), whereas group reallocation requires moving larger tranches for small click deltas. Practitioners should prioritize eliminating demonstrably weak creative–device pairs before fine-tuning keyword-group budgets.

Observational counterfactuals. Simulations assume (i) fixed CPC when scaling spend within a segment, (ii) no auction feedback from policy changes, and (iii) no seasonality beyond May 2024. Real A/B tests or geo experiments would be needed for causal confirmation.

Coverage gaps. Targeting-category analysis (six query types; 83% of budget in “targeted” queries at 12.5% CTR) remains on the backlog. Tablet traffic is sparse and reported only diagnostically. Desktop text has only 14 impressions—unstable CTR.

External validity. Results pertain to one regional bicycle retailer on Yandex Direct. Generalization to other verticals, platforms, or creative pipelines is unknown.

## 7 Conclusion

Structured hypothesis testing on a standard campaign export can separate where budget is wasted from which mechanism drives waste. For Centra Market in May 2024, both segment skew and mobile graphical underperformance are measurable; the latter is sharper, persistent across groups, and addressable with a simple format policy that nearly preserves click volume while improving CTR. Group-level reallocation is directionally correct but yields modest simulated gains, motivating larger or more targeted shifts. We release our analytical protocol as a template for small advertisers lacking custom ML infrastructure but seeking NeurIPS-style rigor in marketing post-mortems.

## References

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